```
> with(linalg) : with(VectorCalculus)
```

[&x, `*`, `+`, `-`, `.`, <,>, <|>, About, AddCoordinates, ArcLength, BasisFormat, Binormal, ConvertVector, CrossProduct, Curl, Curvature, D, Del, DirectionalDiff, Divergence, DotProduct, Flux, GetCoordinateParameters, GetCoordinates, GetNames, GetPVDescription, GetRootPoint, GetSpace, Gradient, Hessian, IsPositionVector, IsRootedVector, IsVectorField, Jacobian, Laplacian, LineInt, MapToBasis, ∇, Norm, Normalize, PathInt, PlotPositionVector, PlotVector, PositionVector, PrincipalNormal, RadiusOfCurvature, RootedVector, ScalarPotential, SetCoordinateParameters, SetCoordinates, SpaceCurve, SurfaceInt, TNBFrame, TangentLine, TangentPlane, TangentVector, Torsion, Vector, VectorField, VectorPotential, VectorSpace, Wronskian, diff, eval, evalVF, int, limit, series] **(1)**

```
> eq1 := 2·x − x^2 − x·y = 0
```

$$eq1 := -x^2 - x\,y + 2\,x = 0 \quad \textbf{(2)}$$

```
> eq2 := −y + x·y = 0
```

$$eq2 := x\,y - y = 0 \quad \textbf{(3)}$$

```
> solve({eq1, eq2}, {x, y})
```

$$\{x = 0, y = 0\}, \{x = 2, y = 0\}, \{x = 1, y = 1\} \quad \textbf{(4)}$$

```
> f1 := (x, y)→2·x − x^2 − x·y
```

$$f1 := (x, y) \mapsto 2\cdot x + (-x^2) + (-x\cdot y) \quad \textbf{(5)}$$

```
> f2 := (x, y)→−y + x·y
```

$$f2 := (x, y) \mapsto -y + x\cdot y \quad \textbf{(6)}$$

```
> Jm := Jacobian([f1(x, y), f2(x, y)], [x, y])
```

$$Jm := \begin{bmatrix} -2\,x - y + 2 & -x \\ y & x - 1 \end{bmatrix} \quad \textbf{(7)}$$

```
> A1 := subs({x = 0, y = 0}, Jm)
```

$$A1 := \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \quad \textbf{(8)}$$

```
> A2 := subs({x = 2, y = 0}, Jm)
```

$$A2 := \begin{bmatrix} -2 & -2 \\ 0 & 1 \end{bmatrix} \quad \textbf{(9)}$$

```
> A3 := subs({x = 1, y = 1}, Jm)
```

(10)

$$A3 := \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \quad (10)$$

```
> eigenvalues(A1)
```

$$2, -1 \quad (11)$$

```
> eigenvalues(A2)
```

$$eigenvalues(A2) \quad (12)$$

```
> eigenvalues(A3)
```

$$-\frac{1}{2} + \frac{I\sqrt{3}}{2}, -\frac{1}{2} - \frac{I\sqrt{3}}{2} \quad (13)$$

```
> with(DEtools)
```

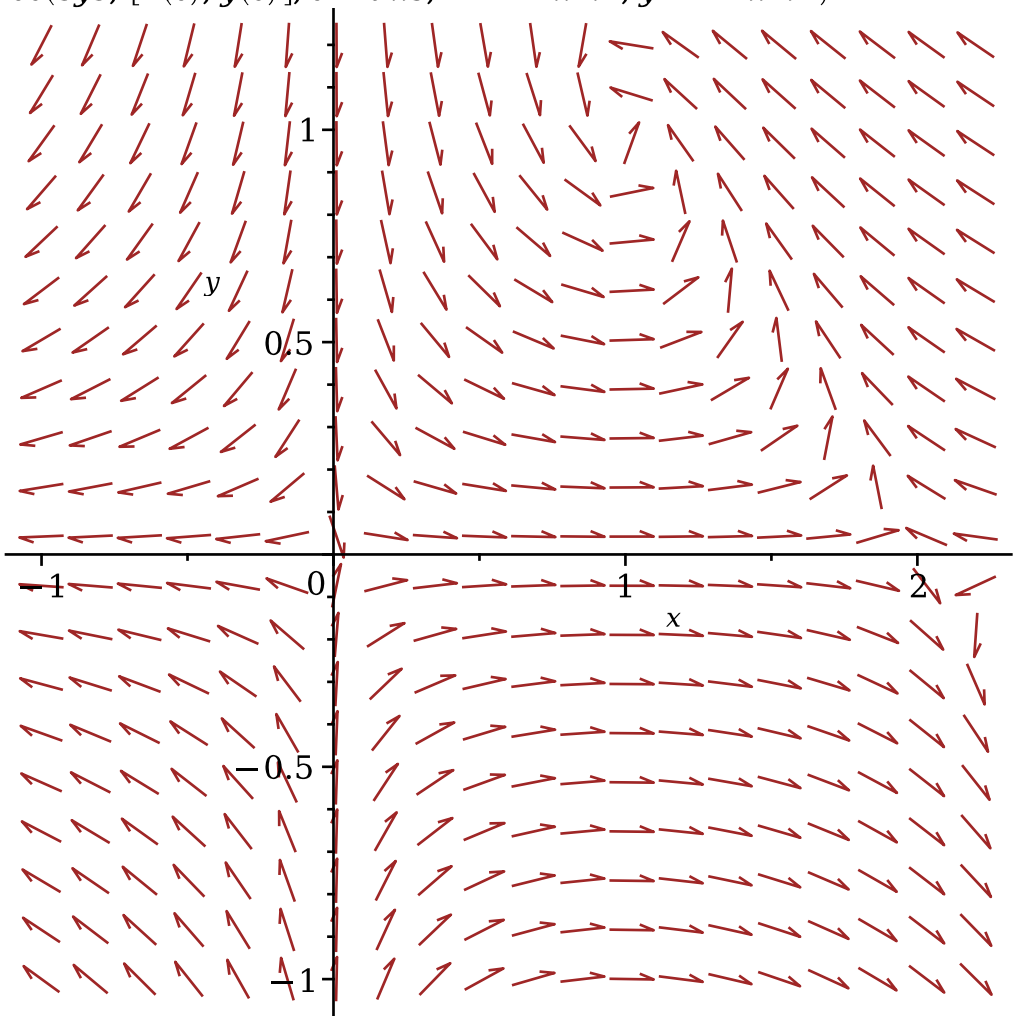
[*AreSimilar*, *Closure*, *DEnormal*, *DEplot*, *DEplot3d*, *DEplot_polygon*, *DFactor*, *DFactorLCLM*, *DFactorsols*, *Dchangevar*, *Desingularize*, *FindODE*, *FunctionDecomposition*, *GCRD*, *Gosper*, *Heunsols*, *Homomorphisms*, *IVPsol*, *IsHyperexponential*, *LCLM*, *MeijerGsols*, *MultiplicativeDecomposition*, *ODEInvariants*, *PDEchangecoords*, *PolynomialNormalForm*, *RationalCanonicalForm*, *ReduceHyperexp*, *RiemannPsols*, *Xchange*, *Xcommutator*, *Xgauge*, *Zeilberger*, *abelsol*, *adjoint*, *autonomous*, *bernoullisol*, *buildsol*, *buildsym*, *canoni*, *caseplot*, *casesplit*, *checkrank*, *chinisol*, *clairautsol*, *constcoeffsols*, *convertAlg*, *convertsys*, *dalembertsol*, *dcoeffs*, *de2diffop*, *dfieldplot*, *diff_table*, *diffop2de*, *dperiodic_sols*, *dpolyform*, *dsubs*, *eigenring*, *endomorphism_charpoly*, *equinv*, *eta_k*, *eulersols*, *exactsol*, *expsols*, *exterior_power*, *firint*, *firtest*, *formal_sol*, *gen_exp*, *generate_ic*, *genhomosol*, *gensys*, *hamilton_eqs*, *hypergeometricsols*, *hypergeomsols*, *hyperode*, *indicialeq*, *infgen*, *initialdata*, *integrate_sols*, *intfactor*, *invariants*, *kovacicsols*, *leftdivision*, *liesol*, *line_int*, *linearsol*, *matrixDE*, *matrix_riccati*, *maxdimsystems*, *moser_reduce*, *muchange*, *mult*, *mutest*, *newton_polygon*, *normalG2*, *ode_int_y*, *ode_y1*, *odeadvisor*, *odepde*, *parametricsol*, *particularsol*, *phaseportrait*, *poincare*, *polysols*, *power_equivalent*, *rational_equivalent*, *ratsols*, *redode*, *reduceOrder*, *reduce_order*, *regular_parts*, *regularsp*, *remove_RootOf*, *riccati_system*, *riccatisol*, *rifread*, *rifsimp*, *rightdivision*, *rtaylor*, *separablesol*, *singularities*, *solve_group*, *super_reduce*, *symgen*, *symmetric_power*, *symmetric_product*, *symtest*, *transinv*, *translate*, *untranslate*, *varparam*, *zoom*] (14)

```
> sys := [diff(x(t), t) = 2·x(t) − x(t)^2 − x(t)·y(t), diff(y(t), t) = −y(t) + x(t)·y(t)]
```

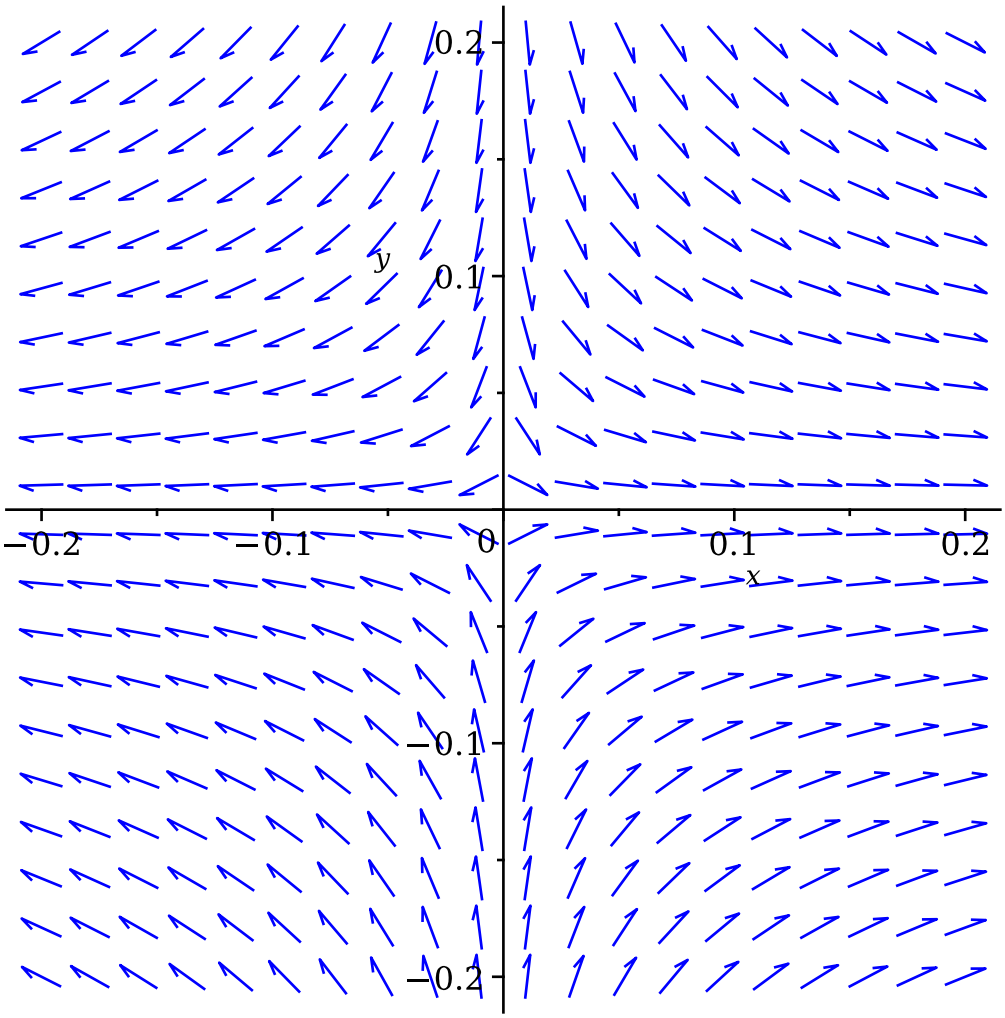
(15)

$$sys := \left[\frac{d}{dt} x(t) = 2\,x(t) - x(t)^2 - x(t)\,y(t),\ \frac{d}{dt} y(t) = -y(t) + x(t)\,y(t) \right] \quad (15)$$

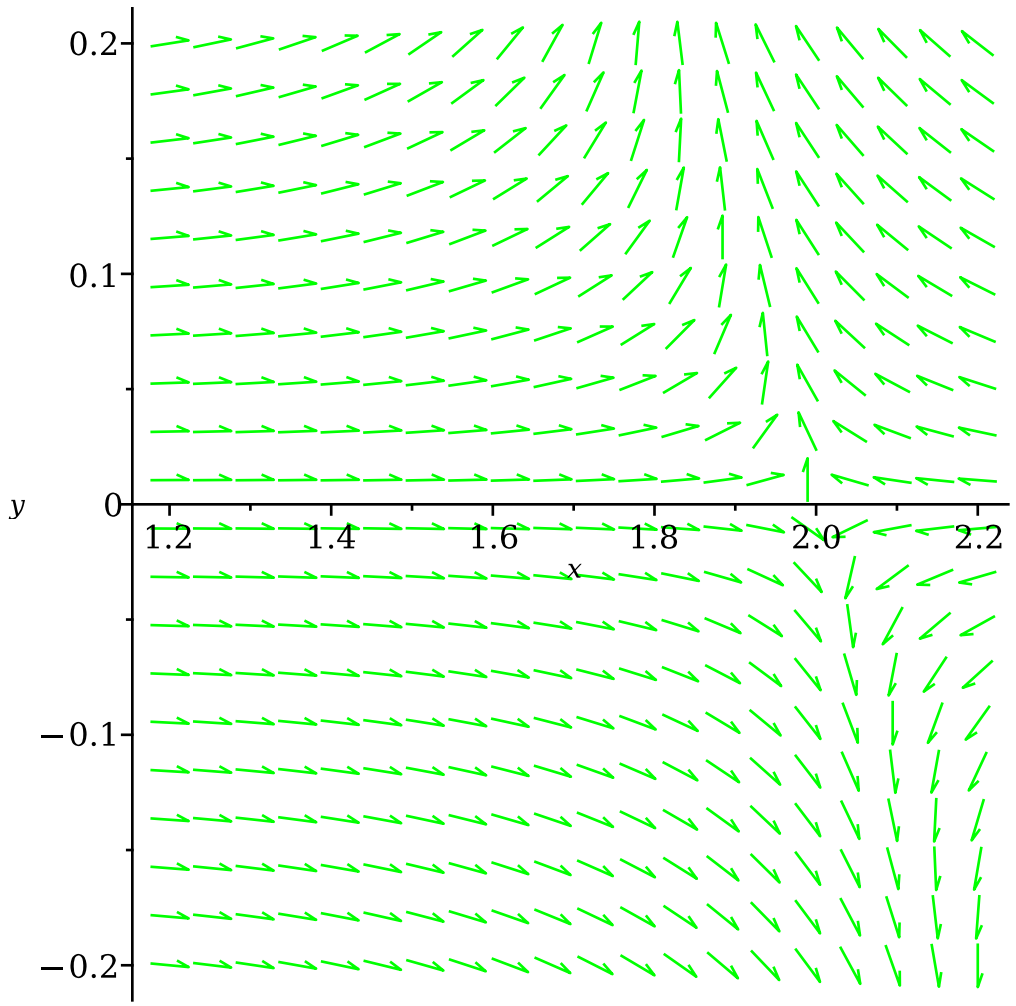
```
> dfieldplot(sys, [x(t), y(t)], t = 0..5, x = -1..2.2, y = -1..1.2)
```



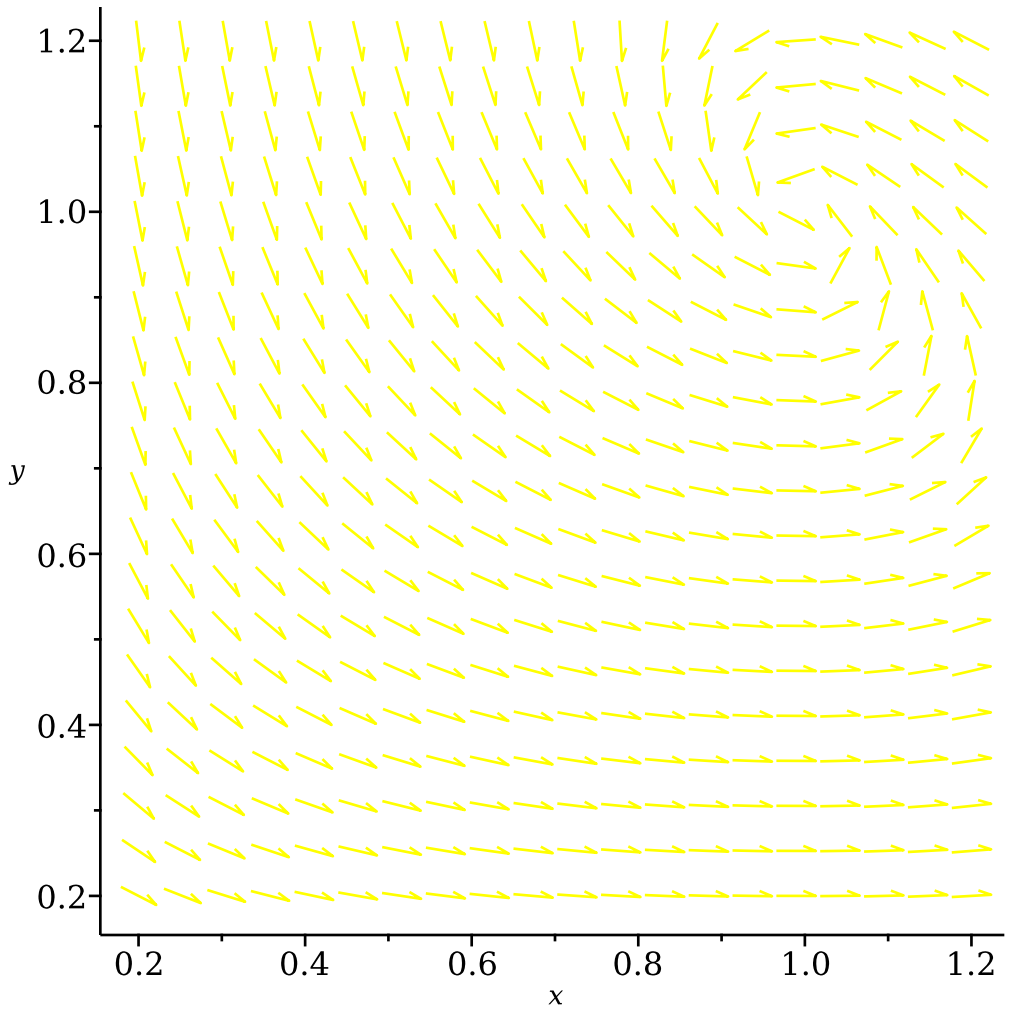

```
> dfieldplot(sys, [x(t), y(t)], t = 0..5, x = -0.2..0.2, y = -0.2..0.2, color = blue)
```



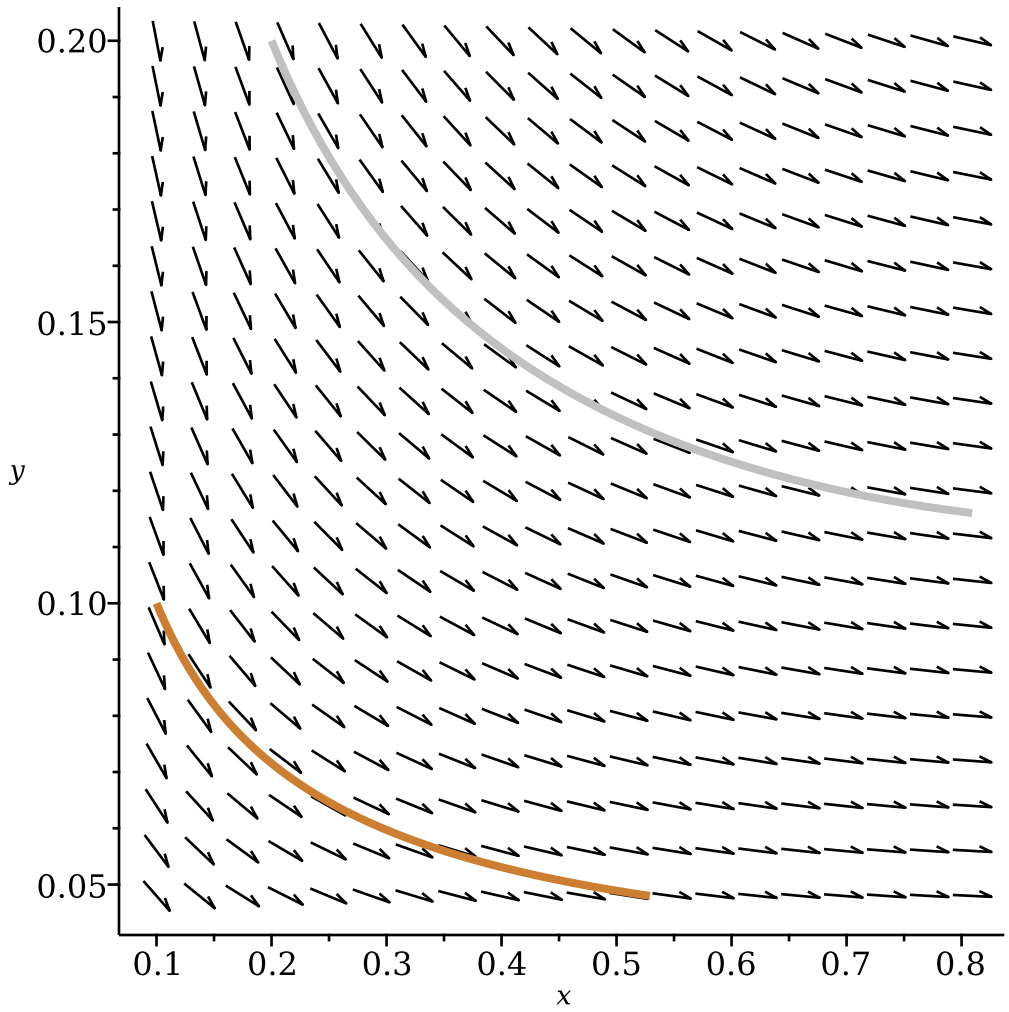

```
> dfieldplot(sys, [x(t), y(t)], t = 0..5, x = 1.2..2.2, y = −0.2..0.2, color = green)
```



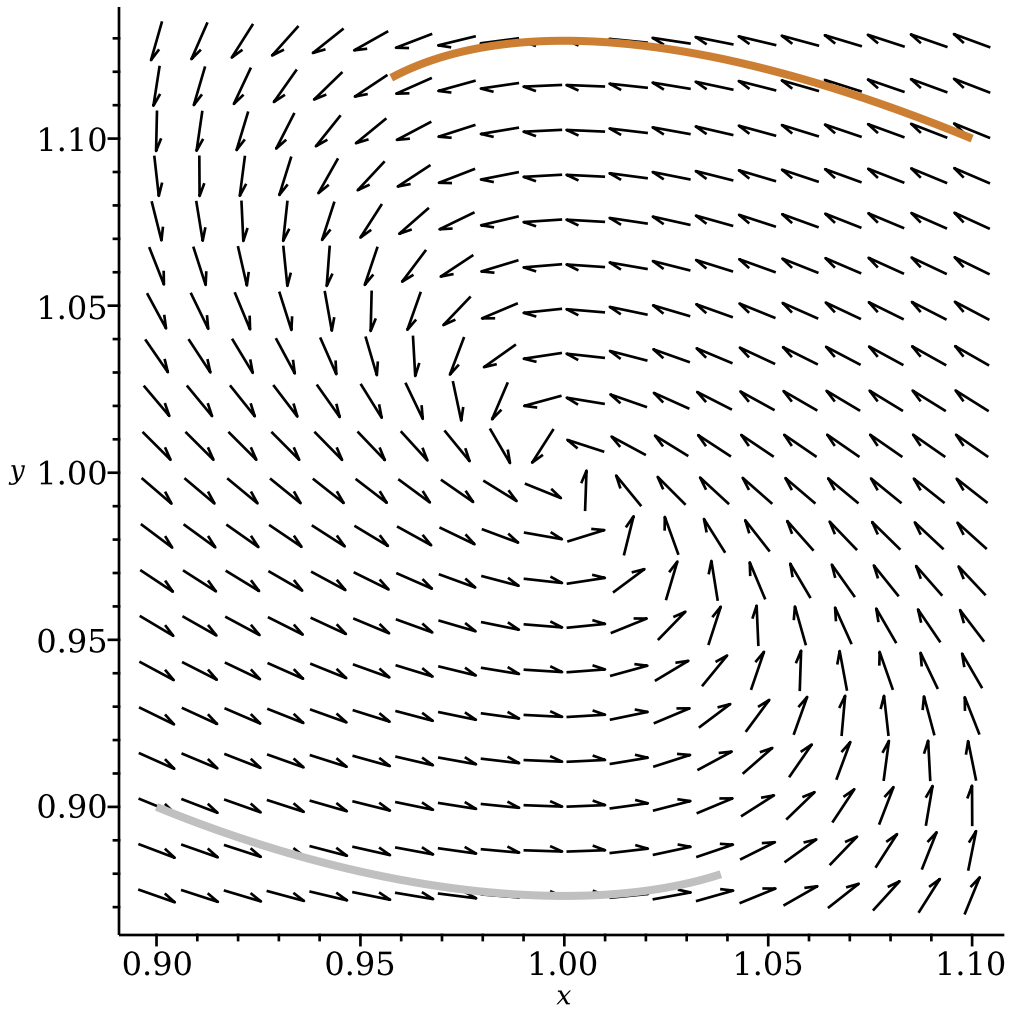

```
> dfieldplot(sys, [x(t), y(t)], t = 0..5, x = 0.2..1.2, y = 0.2..1.2, color = yellow)
```



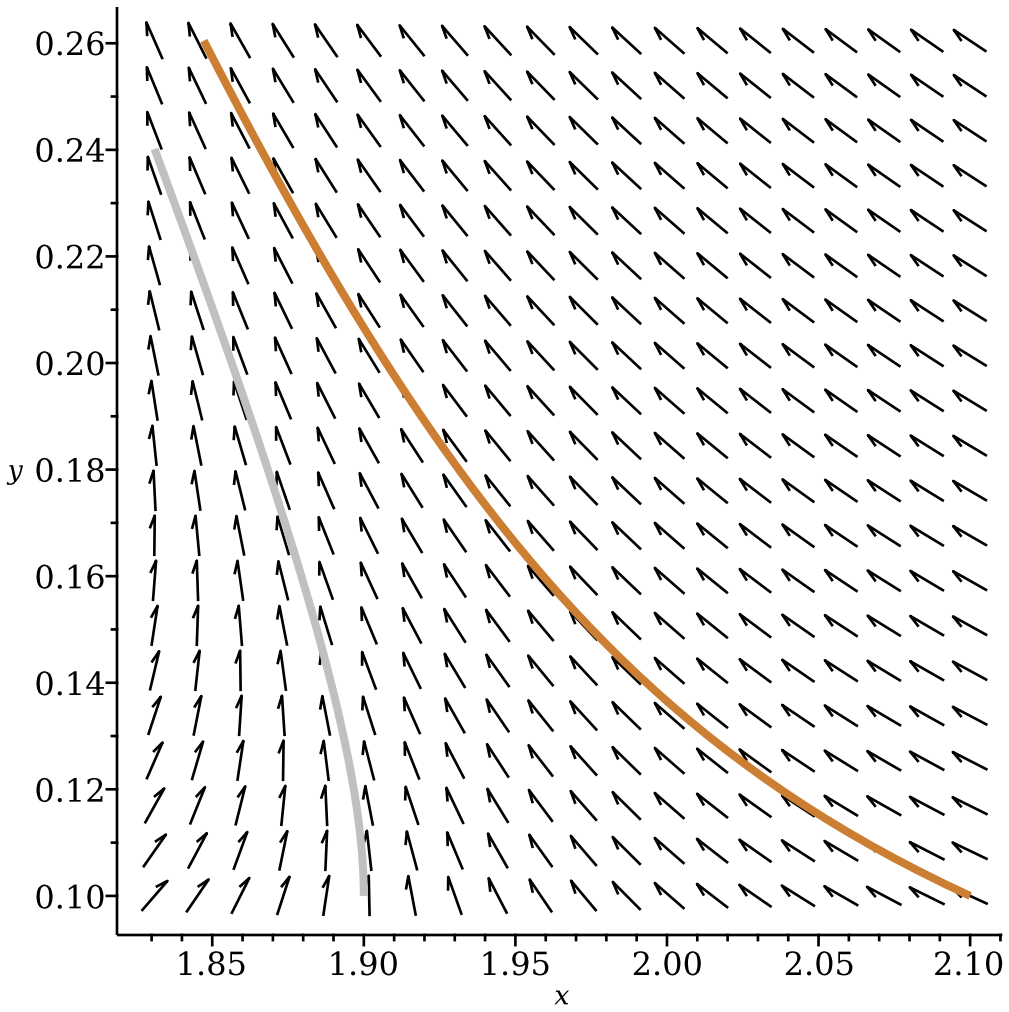

```
> DEplot(sys, [x(t), y(t)], t = 0..1, [[x(0) = 0.2, y(0) = 0.2], [x(0) = 0.1, y(0)
    = 0.1]], linecolor = [silver, gold], color = black)
```



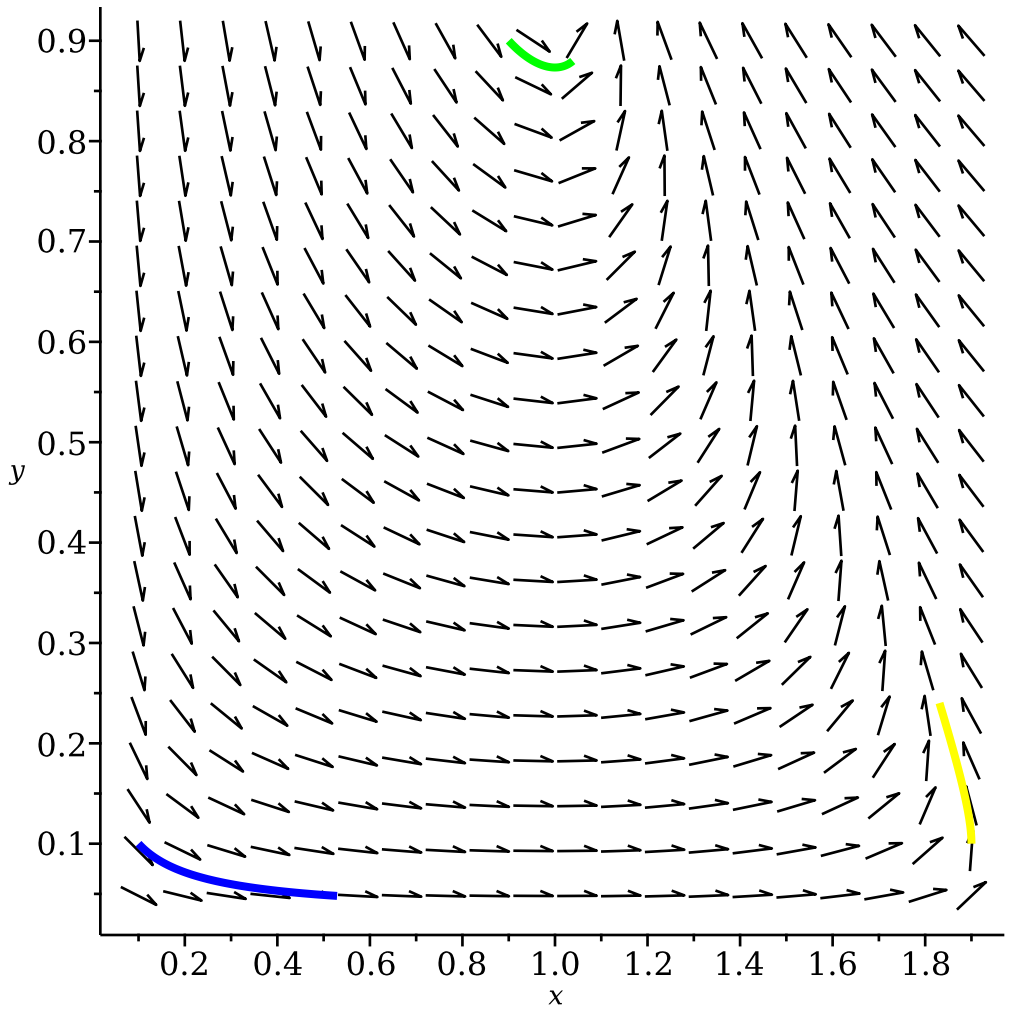

```
> DEplot(sys, [x(t), y(t)], t = 0..1, [[x(0) = 0.9, y(0) = 0.9], [x(0) = 1.1, y(0)
    = 1.1]], linecolor = [silver, gold], color = black)
```



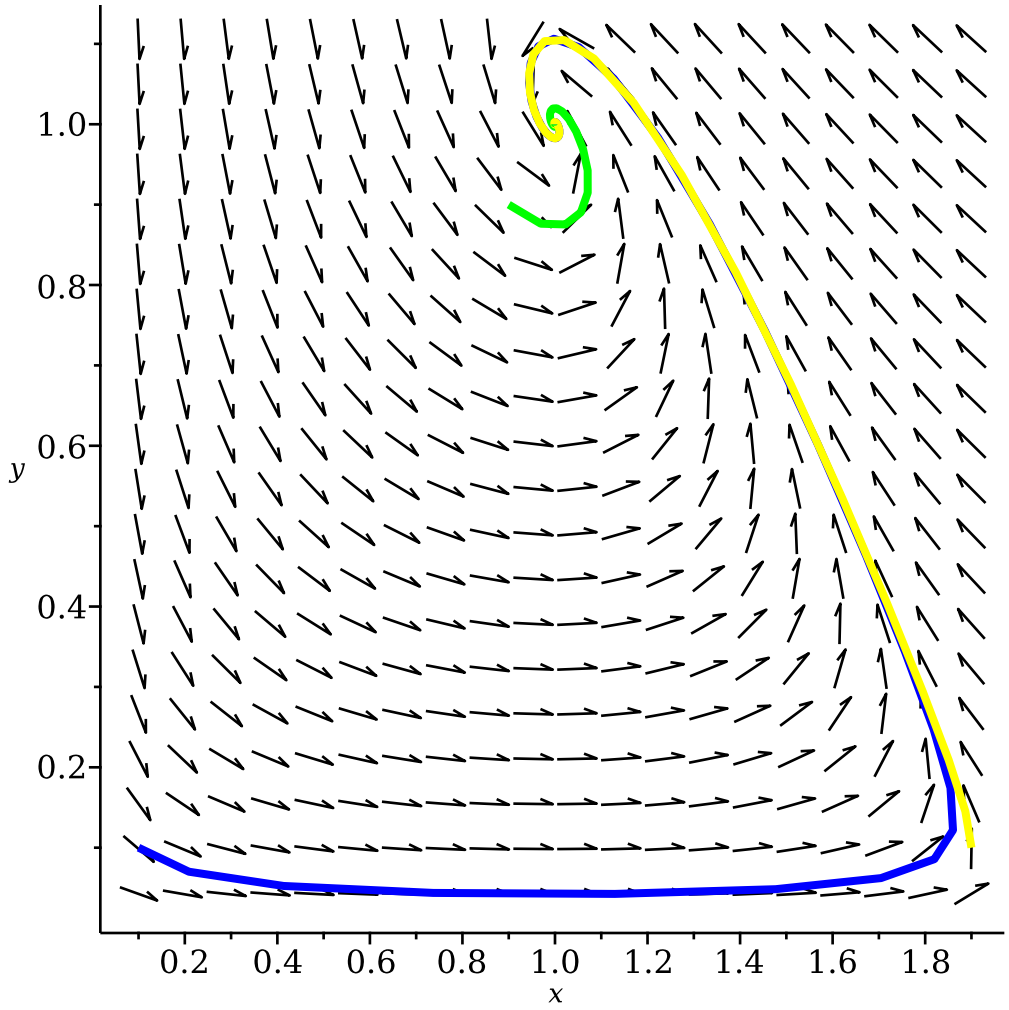

```
> DEplot(sys, [x(t), y(t)], t = 0..1, [[x(0) = 1.9, y(0) = 0.1], [x(0) = 2.1, y(0)
    = 0.1]], linecolor = [silver, gold], color = black)
```




```
> DEplot(sys, [x(t), y(t)], t = 0..1, [[x(0) = 0.1, y(0) = 0.1], [x(0) = 0.9, y(0)
    = 0.9], [x(0) = 1.9, y(0) = 0.1]], linecolor = [blue, green, yellow], color
    = black)
```




```
> DEplot(sys, [x(t), y(t)], t = 0..20, [[x(0) = 0.1, y(0) = 0.1], [x(0) = 0.9, y(0) = 0.9], [x(0) = 1.9, y(0) = 0.1]], linecolor = [blue, green, yellow], color = black)
```



> $eq1 := x - 2 \cdot x \cdot y = 0$

$$eq1 := -2\,x\,y + x = 0 \quad (16)$$

> $eq2 := \frac{x^2}{2} - y = 0$

$$eq2 := \frac{x^2}{2} - y = 0 \quad (17)$$

> $solve(\{eq1, eq2\}, \{x, y\})$

$$\{x = 0, y = 0\}, \left\{x = 1, y = \frac{1}{2}\right\}, \left\{x = -1, y = \frac{1}{2}\right\} \quad (18)$$

> $with(linalg) : with(VectorCalculus)$

[&x, `*`, `+`, `-`, `.`, <,>, <|>, About, AddCoordinates, ArcLength, (19)
BasisFormat, Binormal, ConvertVector, CrossProduct, Curl, Curvature, D,
Del, DirectionalDiff, Divergence, DotProduct, Flux,
GetCoordinateParameters, GetCoordinates, GetNames,
GetPVDescription, GetRootPoint, GetSpace, Gradient, Hessian,
IsPositionVector, IsRootedVector, IsVectorField, Jacobian, Laplacian,
LineInt, MapToBasis,∇, Norm, Normalize, PathInt, PlotPositionVector,

PlotVector, PositionVector, PrincipalNormal, RadiusOfCurvature, RootedVector, ScalarPotential, SetCoordinateParameters, SetCoordinates, SpaceCurve, SurfaceInt, TNBFrame, TangentLine, TangentPlane, TangentVector, Torsion, Vector, VectorField, VectorPotential, VectorSpace, Wronskian, diff, eval, evalVF, int, limit, series]

> $f1 := (x, y) \to x - 2*x*y$

$$f1 := (x, y) \mapsto x + (-2 \cdot x \cdot y) \quad (20)$$

> $f2 := (x, y) \to \frac{x^2}{2} - y$

$$f2 := (x, y) \mapsto x^2 \cdot 2^{-1} + (-y) \quad (21)$$

> $J := Jacobian([f1(x, y), f2(x, y)], [x, y])$

$$J := \begin{bmatrix} -2y + 1 & -2x \\ x & -1 \end{bmatrix} \quad (22)$$

> $A1 := subs(\{x = 0, y = 0\}, J)$

$$A1 := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (23)$$

> $A2 := subs\left(\left\{x = 1, y = \frac{1}{2}\right\}, J\right)$

$$A2 := \begin{bmatrix} 0 & -2 \\ 1 & -1 \end{bmatrix} \quad (24)$$

> $A3 := subs\left(\left\{x = -1, y = \frac{1}{2}\right\}, J\right)$

$$A3 := \begin{bmatrix} 0 & 2 \\ -1 & -1 \end{bmatrix} \quad (25)$$

> $eigenvalues(A1)$

$$1, -1 \quad (26)$$

> $eigenvalues(A2)$

$$-\frac{1}{2} + \frac{I\sqrt{7}}{2}, -\frac{1}{2} - \frac{I\sqrt{7}}{2} \quad (27)$$

> $eigenvalues(A3)$

$$-\frac{1}{2} + \frac{I\sqrt{7}}{2}, -\frac{1}{2} - \frac{I\sqrt{7}}{2} \quad (28)$$

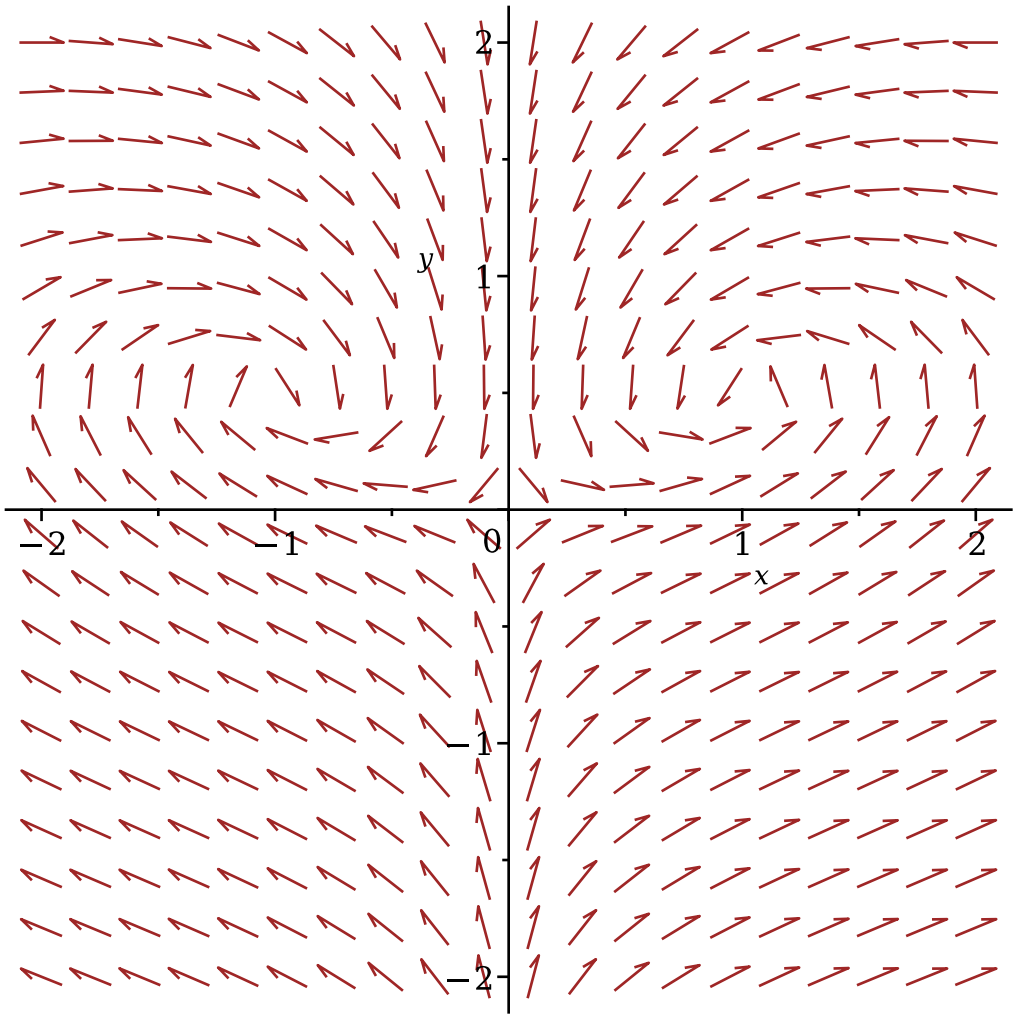
> $with(DEtools)$

[*AreSimilar, Closure, DEnormal, DEplot, DEplot3d, DEplot_polygon, DFactor, DFactorLCLM, DFactorsols, Dchangevar, Desingularize, FindODE, FunctionDecomposition, GCRD, Gosper, Heunsols, Homomorphisms, IVPsol, IsHyperexponential, LCLM, MeijerGsols, MultiplicativeDecomposition, ODEInvariants, PDEchangecoords, PolynomialNormalForm, RationalCanonicalForm, ReduceHyperexp, RiemannPsols, Xchange, Xcommutator, Xgauge, Zeilberger, abelsol, adjoint, autonomous, bernoullisol, buildsol, buildsym, canoni, caseplot, casesplit, checkrank, chinisol, clairautsol, constcoeffsols, convertAlg, convertsys, dalembertsol, dcoeffs, de2diffop, dfieldplot, diff_table, diffop2de, dperiodic_sols, dpolyform, dsubs, eigenring, endomorphism_charpoly, equinv, eta_k, eulersols, exactsol, expsols, exterior_power, firint, firtest, formal_sol, gen_exp, generate_ic, genhomosol, gensys, hamilton_eqs, hypergeometricsols, hypergeomsols, hyperode, indicialeq, infgen, initialdata, integrate_sols, intfactor, invariants, kovacicsols, leftdivision, liesol, line_int, linearsol, matrixDE, matrix_riccati, maxdimsystems, moser_reduce, muchange, mult, mutest, newton_polygon, normalG2, ode_int_y, ode_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power_equivalent, rational_equivalent, ratsols, redode, reduceOrder, reduce_order, regular_parts, regularsp, remove_RootOf, riccati_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve_group, super_reduce, symgen, symmetric_power, symmetric_product, symtest, transinv, translate, untranslate, varparam, zoom*] **(29)**

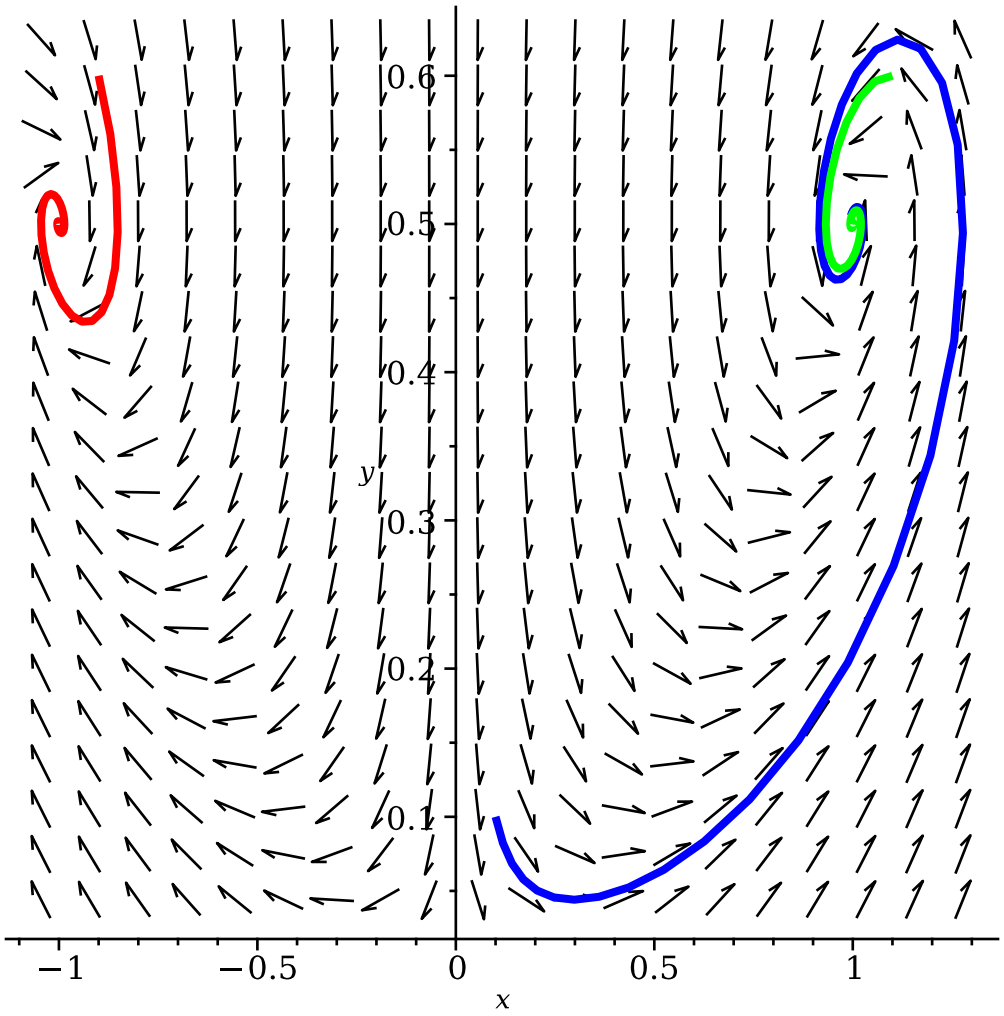
> $sys := [diff(x(t),t) = x(t) - 2*x(t)*y(t), diff(y(t),t) = (1/2)*x(t)^2 - y(t)]$

$$sys := \left[\frac{d}{dt}x(t) = x(t) - 2\,x(t)\,y(t),\ \frac{d}{dt}y(t) = \frac{x(t)^2}{2} - y(t)\right] \quad \textbf{(30)}$$

> $dfieldplot(sys, [x(t), y(t)], t = 0..5, x = -2..2, y = -2..2)$



> $DEplot(sys, [x(t), y(t)], t = 0..10, [[x(0) = 0.1, y(0) = 0.1], [x(0) = 1.1, y(0) = 6/10], [x(0) = -0.9, y(0) = 6/10]], linecolor = [blue, green, red], color = black)$



> $eq1 := y = 0$

$$eq1 := y = 0 \quad (31)$$

> $eq2 := -4 \cdot \sin(x) = 0$

$$eq2 := -4\sin(x) = 0 \quad (32)$$

> $solve(\{eq1, eq2\}, \{x, y\})$

$$\{x = 0, y = 0\} \quad (33)$$

> $with(linalg) : with(VectorCalculus)$

[&x, `*`, `+`, `-`, `.`, <,>, <|>, *About*, *AddCoordinates*, *ArcLength*, *BasisFormat*, *Binormal*, *ConvertVector*, *CrossProduct*, *Curl*, *Curvature*, D, *Del*, *DirectionalDiff*, *Divergence*, *DotProduct*, *Flux*, *GetCoordinateParameters*, *GetCoordinates*, *GetNames*, *GetPVDescription*, *GetRootPoint*, *GetSpace*, *Gradient*, *Hessian*, *IsPositionVector*, *IsRootedVector*, *IsVectorField*, *Jacobian*, *Laplacian*, *LineInt*, *MapToBasis*, ∇, *Norm*, *Normalize*, *PathInt*, *PlotPositionVector*, *PlotVector*, *PositionVector*, *PrincipalNormal*, *RadiusOfCurvature*, *RootedVector*, *ScalarPotential*, *SetCoordinateParameters*, *SetCoordinates*, *SpaceCurve*, *SurfaceInt*, *TNBFrame*, *TangentLine*, (34)

TangentPlane, TangentVector, Torsion, Vector, VectorField, VectorPotential, VectorSpace, Wronskian, diff, eval, evalVF, int, limit, series]

> $f1 := (x, y) \to y$

$$f1 := (x, y) \mapsto y \tag{35}$$

> $f2 := (x, y) \to -4 \cdot \sin(x)$

$$f2 := (x, y) \mapsto -4 \cdot \sin(x) \tag{36}$$

> $J := Jacobian([f1(x, y), f2(x, y)], [x, y])$

$$J := \begin{bmatrix} 0 & 1 \\ -4\cos(x) & 0 \end{bmatrix} \tag{37}$$

> $A := subs([x = 0, y = 0], J)$

$$A := \begin{bmatrix} 0 & 1 \\ -4\cos(0) & 0 \end{bmatrix} \tag{38}$$

> $eigenvalues(A)$

$$2\,\mathrm{I}, -2\,\mathrm{I} \tag{39}$$

> $eq := diff(y(x), x) = -\dfrac{4 \cdot \sin(x)}{y(x)}$

$$eq := \frac{\mathrm{d}}{\mathrm{d}x} y(x) = -\frac{4\sin(x)}{y(x)} \tag{40}$$

> $sol := dsolve(eq, y(x))$

$$sol := y(x) = \sqrt{8\cos(x) + c_1}, y(x) = -\sqrt{8\cos(x) + c_1} \tag{41}$$

> $H := (x, y) \to y^2 - 8 \cdot \cos(x)$

$$H := (x, y) \mapsto y^2 + (-8 \cdot \cos(x)) \tag{42}$$

> $eqH := diff(H(x, y), x) \cdot y - 4 \cdot diff(H(x, y), y) \cdot \sin(x)$

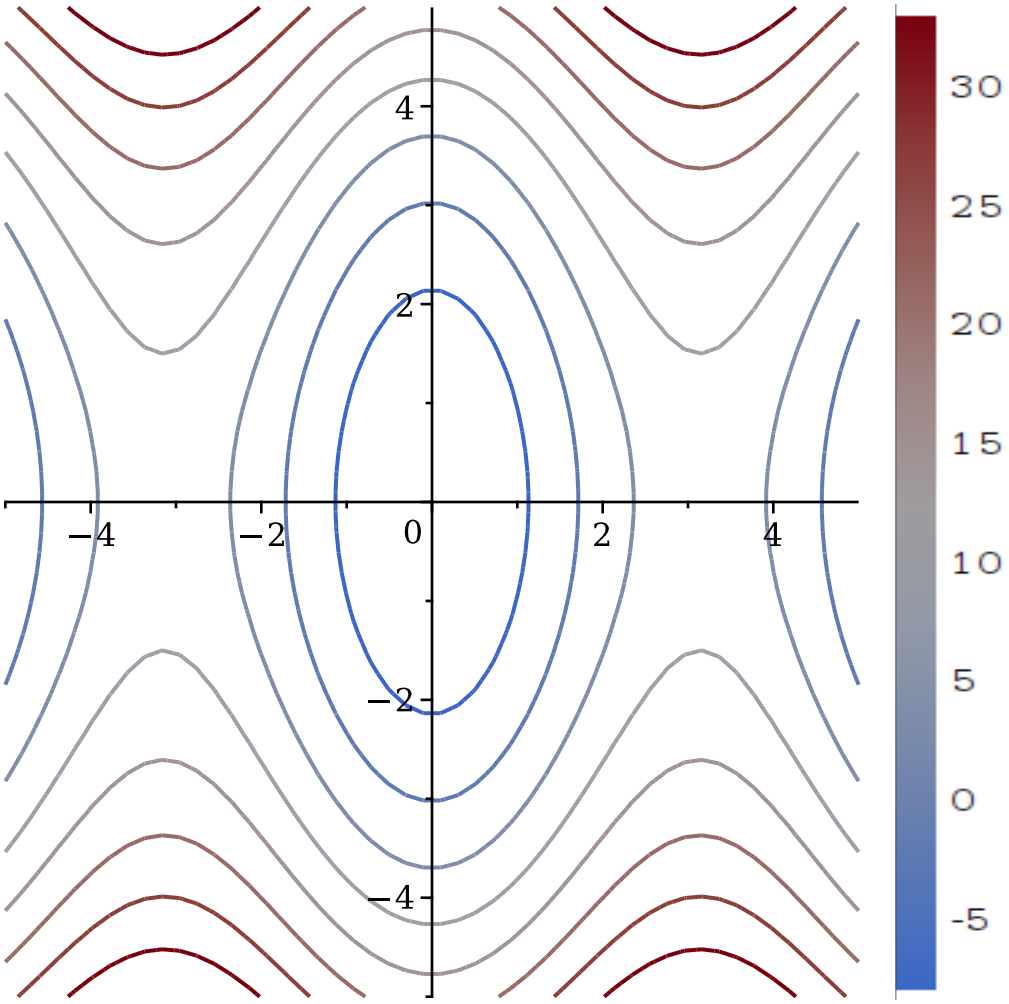
$$eqH := 0 \tag{43}$$

> $with(plots)$

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot,* (44)

textplot3d, tubeplot]

> *contourplot*($H(x, y)$, $x = -5..5$, $y = -5..5$)



> $eq1 := x - x \cdot y = 0$

$$eq1 := -x\,y + x = 0 \quad (45)$$

> $eq2 := -0.3 \cdot y + 0.3 \cdot x \cdot y = 0$

$$eq2 := -0.3\,y + 0.3\,x\,y = 0 \quad (46)$$

> *solve*($\{eq1, eq2\}$, $\{x, y\}$)

$$\{x = 0., y = 0.\}, \{x = 1., y = 1.\} \quad (47)$$

> $f1 := (x, y) \rightarrow x - x \cdot y$

$$f1 := (x, y) \mapsto x + (-x \cdot y) \quad (48)$$

> $f2 := (x, y) \rightarrow -0.3 \cdot y + 0.3 \cdot x \cdot y$

$$f2 := (x, y) \mapsto -0.3 \cdot y + 0.3 \cdot x \cdot y \quad (49)$$

> $J := Jacobian([f1(x, y), f2(x, y)], [x, y])$

$$J := \begin{bmatrix} -y + 1 & -x \\ 0.3\,y & -0.3 + 0.3\,x \end{bmatrix} \quad (50)$$

> $A1 := subs([x = 1, y = 1], J)$

$$A1 := \begin{bmatrix} 0 & -1 \\ 0.3 & 0. \end{bmatrix} \quad (51)$$

> $eigenvalues(A1)$

$$0. + 0.547722557505166\,I,\ 0. - 0.547722557505166\,I \quad (52)$$

> $H := (x, y) \to y - \ln(y) + 0.3 \cdot (x - \ln(x))$

$$H := (x, y) \mapsto y + (-\ln(y)) + 0.3 \cdot (x + (-\ln(x))) \quad (53)$$

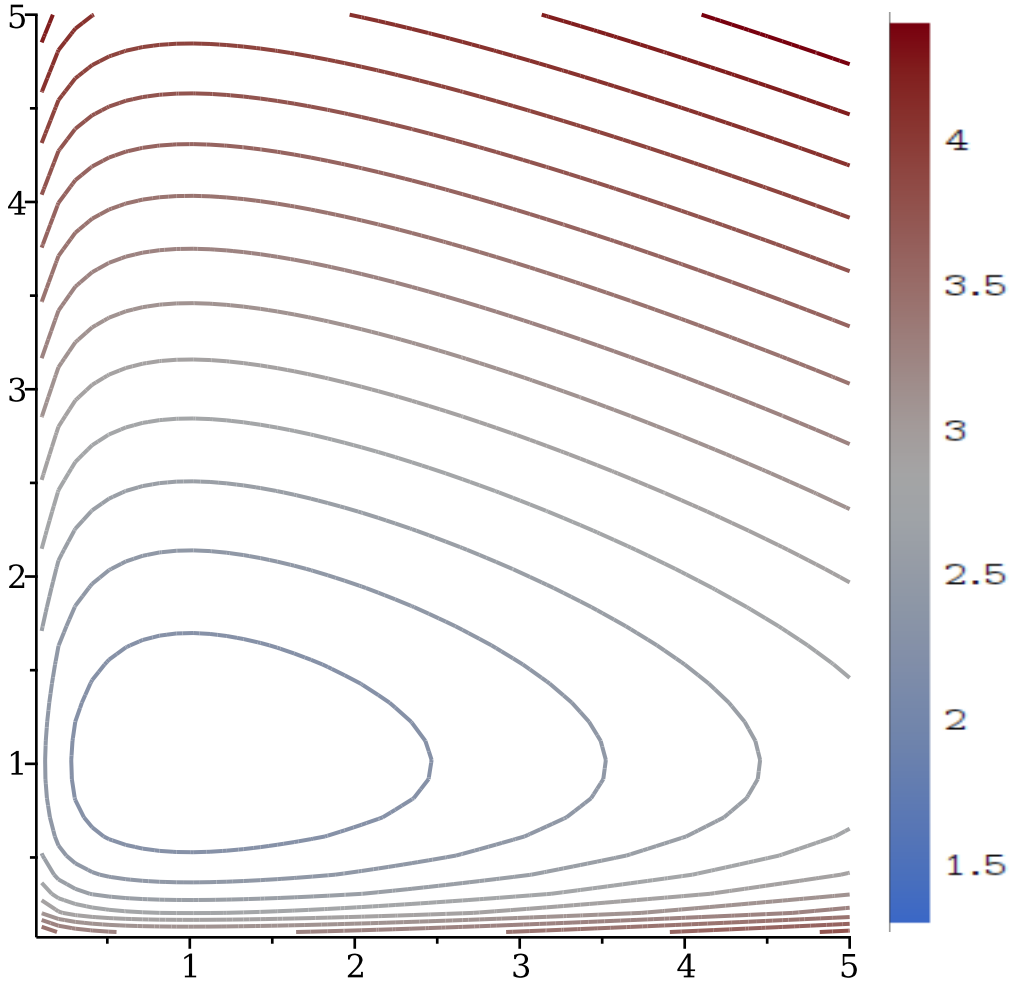
> $eqH := diff(H(x, y), x) \cdot (x - x \cdot y) + diff(H(x, y), y) \cdot (-0.3 \cdot y + 0.3 \cdot x \cdot y)$

$$eqH := \left(0.3 - \frac{0.3}{x}\right)(-x\,y + x) + \left(1 - \frac{1}{y}\right)(-0.3\,y + 0.3\,x\,y) \quad (54)$$

> $simplify(\%)$

$$0. \quad (55)$$

> $contourplot(H(x, y), x = 0..5, y = 0..5, contours = 20)$



>